

Origins and Evolution of Language

Class 1: Evolutionary processes shaping language and the capacity for language

Kenny Smith

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Communication is widespread, but language is unique

How did language evolve?

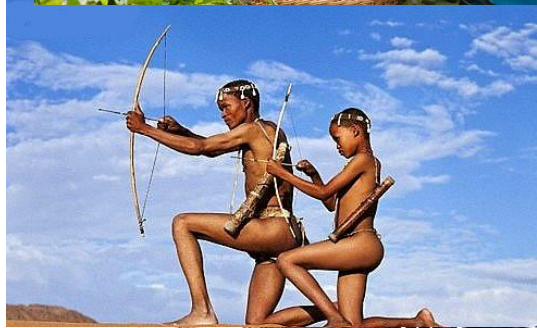


Language is universal in our species

Language is a hugely **adaptive** trait



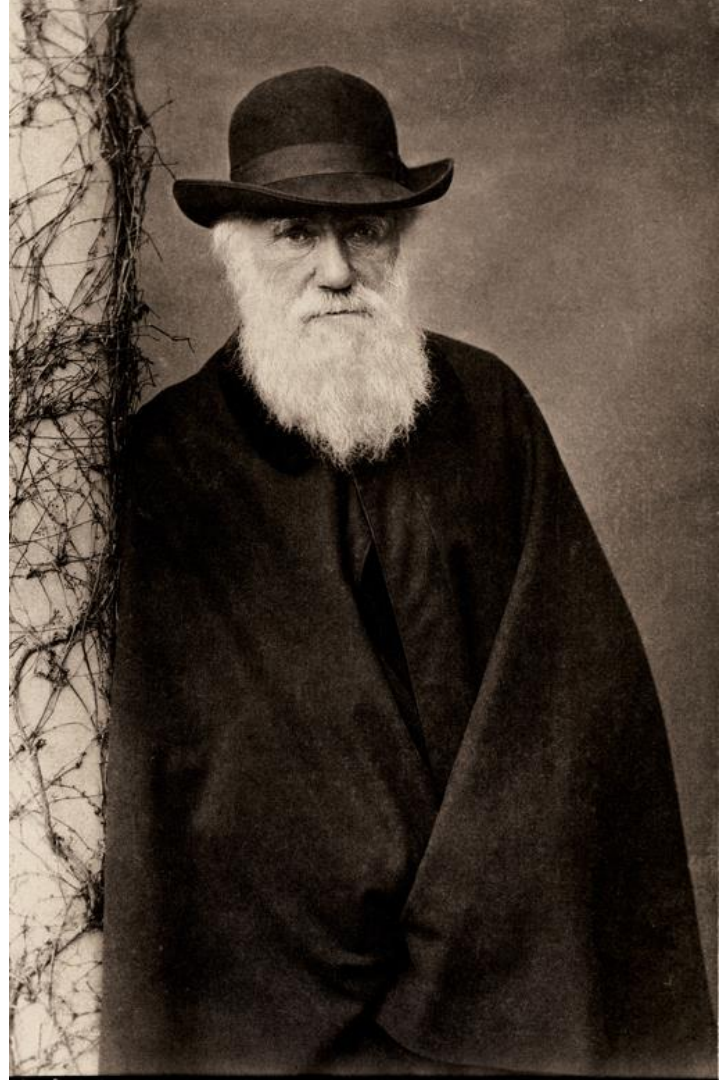
A tool for the communication of knowledge and internal states



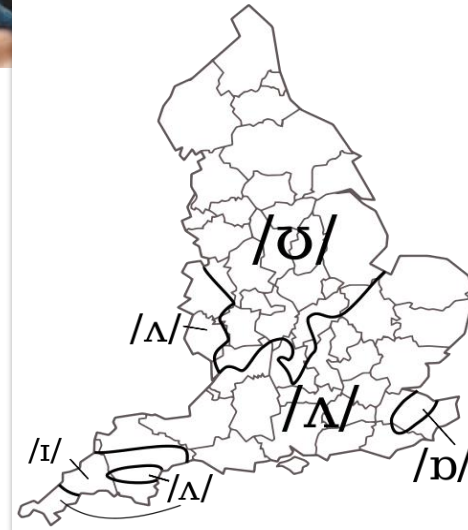
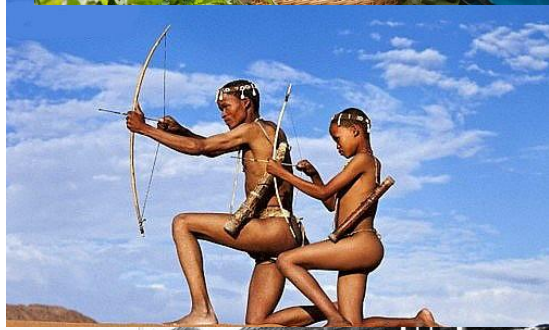
One possible explanation

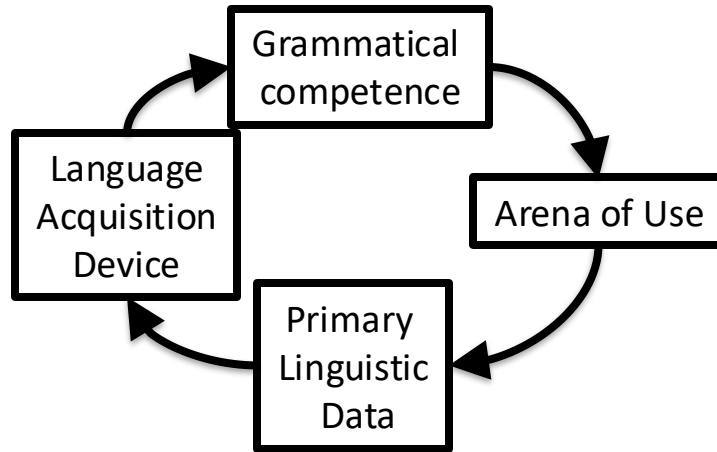
- Language is just like any other adaptive feature of an organism's biology
- It's an **innate** feature of the human mind
- It evolved by natural selection under pressure for communication

More on this shortly



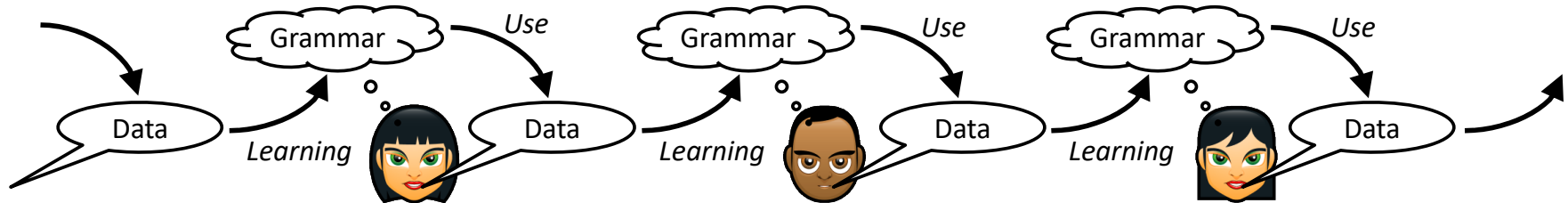
Social learning is ubiquitous in humans





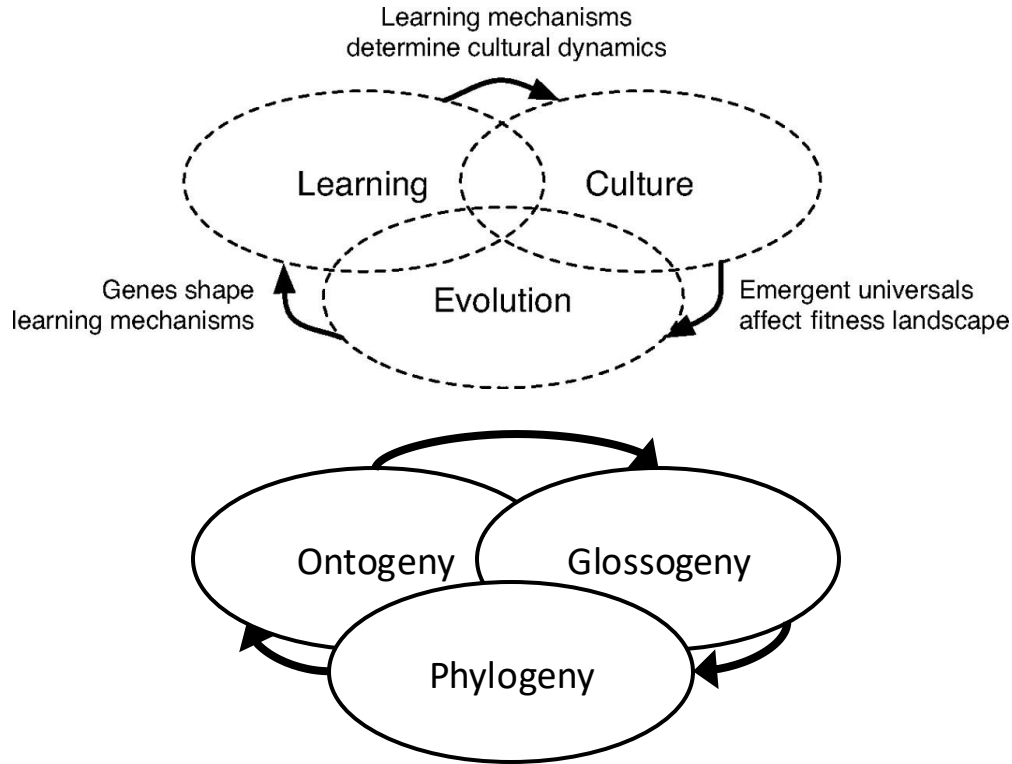
Language is transmitted
via repeated **learning**
and **use**

Language is shaped as a
consequence of these
processes



Upper: from Hurford, J. R. (1990). Nativist and functional explanations in language acquisition. In I. M. Roca (Ed.), *Logical issues in language acquisition* (pp. 85–136). Dordrecht: Foris.
Lower: from Smith, K. (2022). How language learning and language use create linguistic structure. *Current Directions in Psychological Science*, 31, 177-186.

Learning, **culture** and biology



Upper: from Kirby, S., Dowman, M., & Griffiths, T. (2007). Innateness and culture in the evolution of language. *PNAS*, 104, 5241-5245.

Lower: adapted from Fitch, W. T. (2010). *The Evolution of Language*. Cambridge: Cambridge University Press

Quick bit of admin

<https://kennysmithed.github.io/origins25lsa/>

Course outline

Class	Topics
1	Natural selection, very basics of human evolution, language as a biological adaptation, the alternative
2-3	Linguistic structure as a cultural adaptation to pressures from learning and use
4	Cognitive prerequisites, comparative psychology of those prerequisites
5	Gene-culture co-evolution



Pre-lecture preparation

- **Readings should normally be done in advance**
- Do the reading, answer the quiz questions on Canvas
 - Most useful bit for me is the free comment box at the end
- I will assume you have done the readings, we'll talk about them in class

Assessment

Grades for the course will be based on the pre-lecture quizzes

Show of hands: who actually needs/wants a grade?

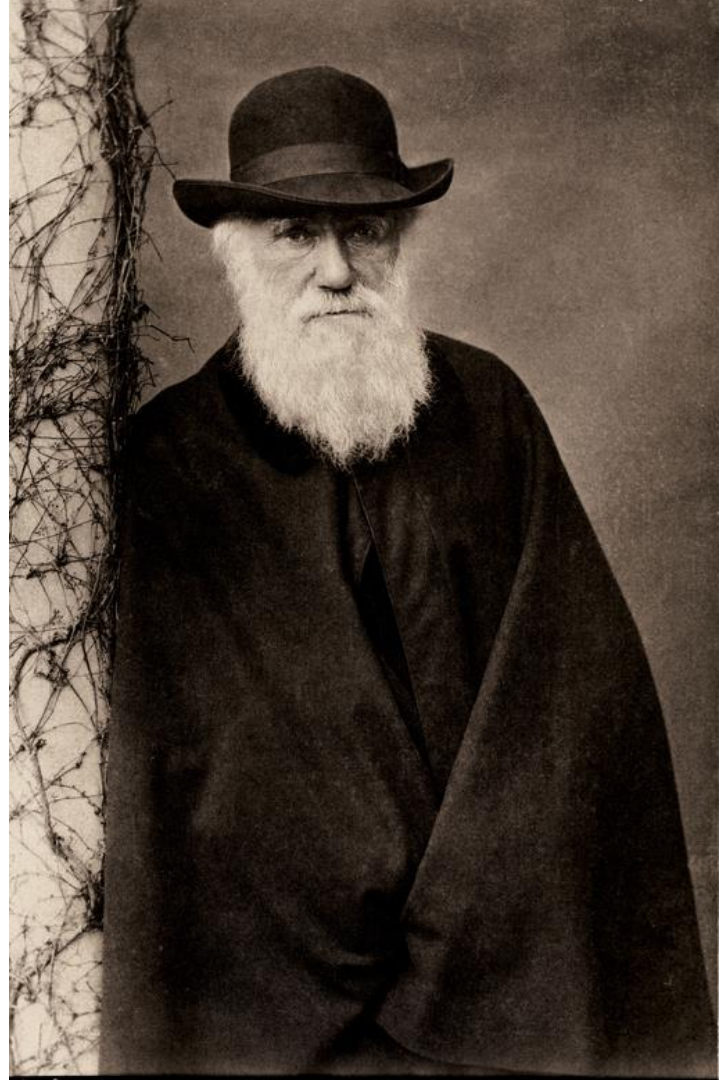
Outline for the rest of today

- Primer on natural selection and **adaptation**
- Language as an adaptation
- Very brief basics of human evolution

Primer on natural selection and adaptation

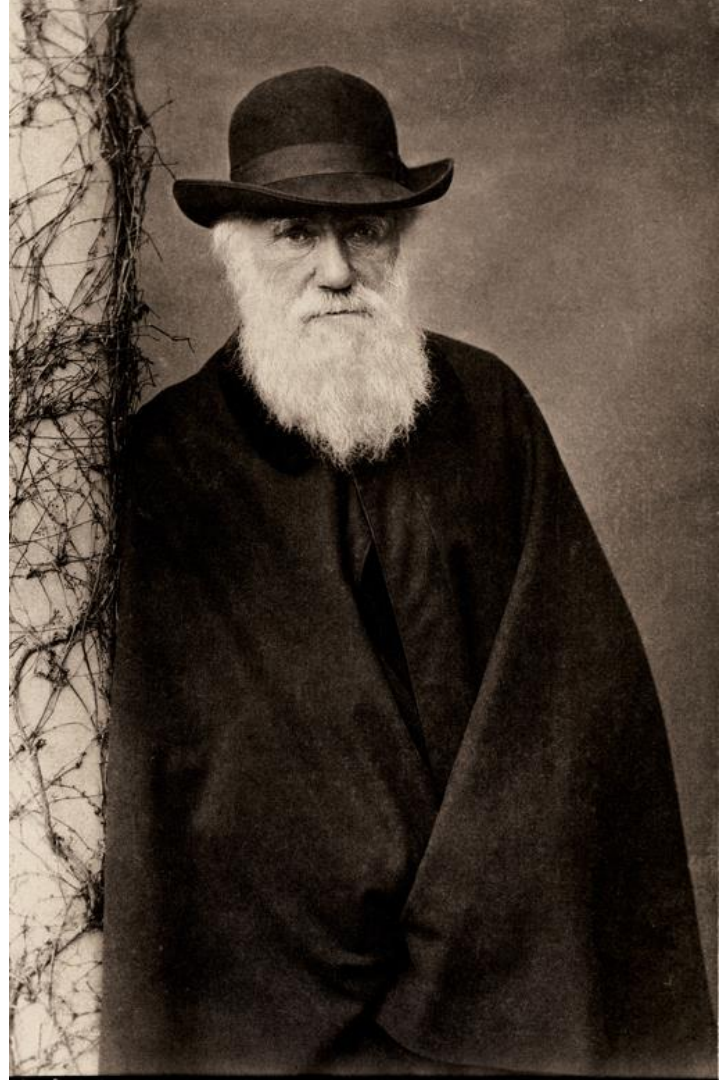
One possible explanation

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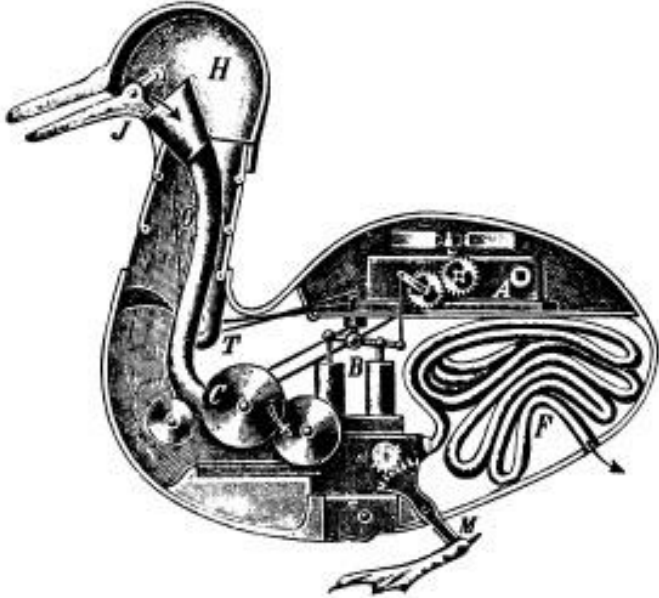


One possible explanation

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Design in life



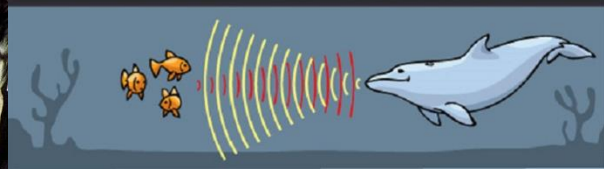
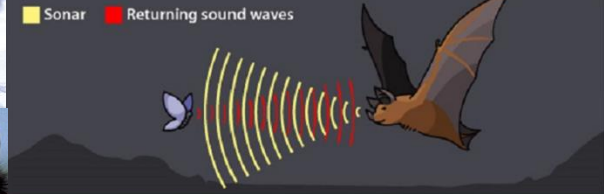
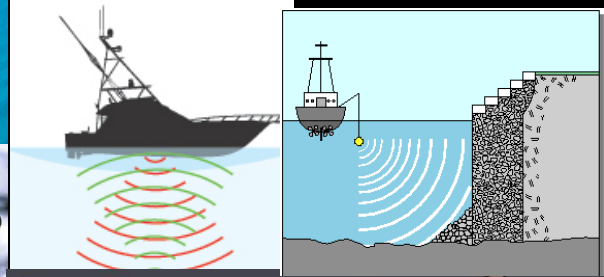
https://en.wikipedia.org/wiki/Digesting_Duck

Living things are remarkably competent

- They are self-sustaining, reproducing, autonomous, deal with hostile, changing environments, ...
- Designing a machine to do any of these things is currently beyond us



Design in life



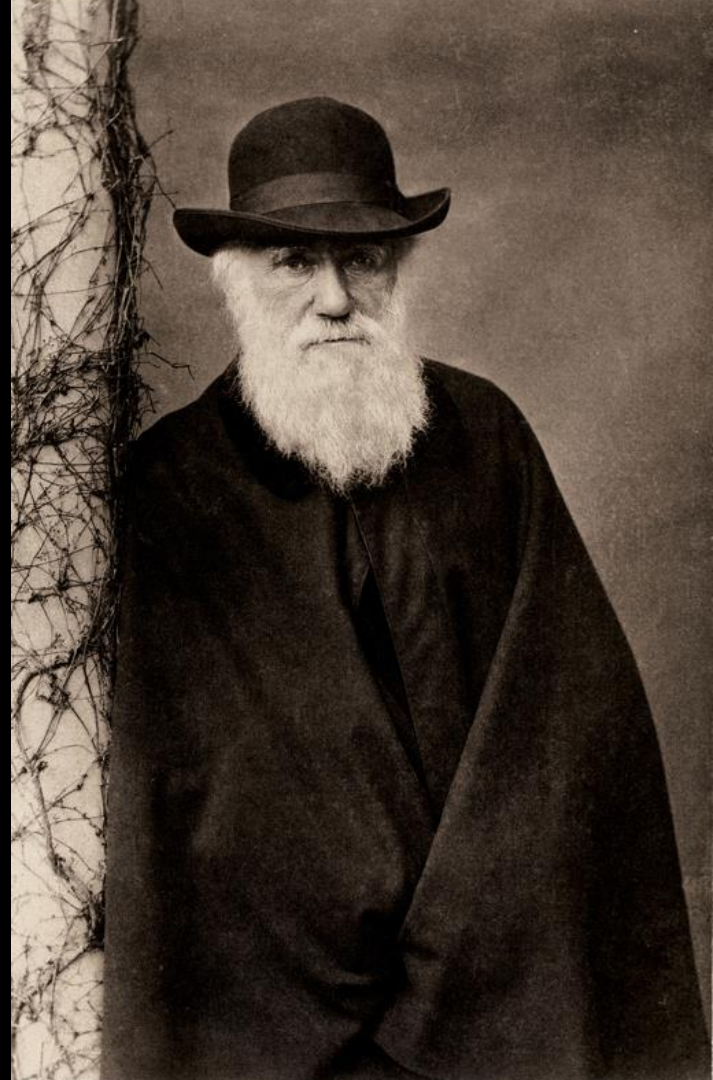
Adaptations

- “‘design’ in life – those properties of living things that enable them to survive and reproduce in nature.” (Ridley, 1996, *Evolution*, p. 5)
- “An adaptation is a characteristic that enhances the survival or reproduction of organisms that bear it” (Futuyama, 2009, *Evolution*, p. 279)
- “The process whereby the members of a population become better suited to some feature of their environment through change in a characteristic that affects their survival and reproduction” (Futuyama again, same page)
- “a characteristic that has evolved by natural selection” (Futuyama again, same page)

Evolution by natural selection

“Owing to this struggle for life, any variation, however slight and from whatever cause proceeding, if it be in any degree profitable to an individual of any species, in its infinitely complex relations to other organic beings and to external nature, will tend to the preservation of that individual, and will generally be inherited by its offspring. The offspring, also, will thus have a better chance of surviving, for, of the many individuals of any species which are periodically born, but a small number can survive. I have called this principle, by which each slight variation, if useful, is preserved, by the term of Natural Selection”

(Darwin, 1859/1965, *On the Origin of Species*, p. 61)

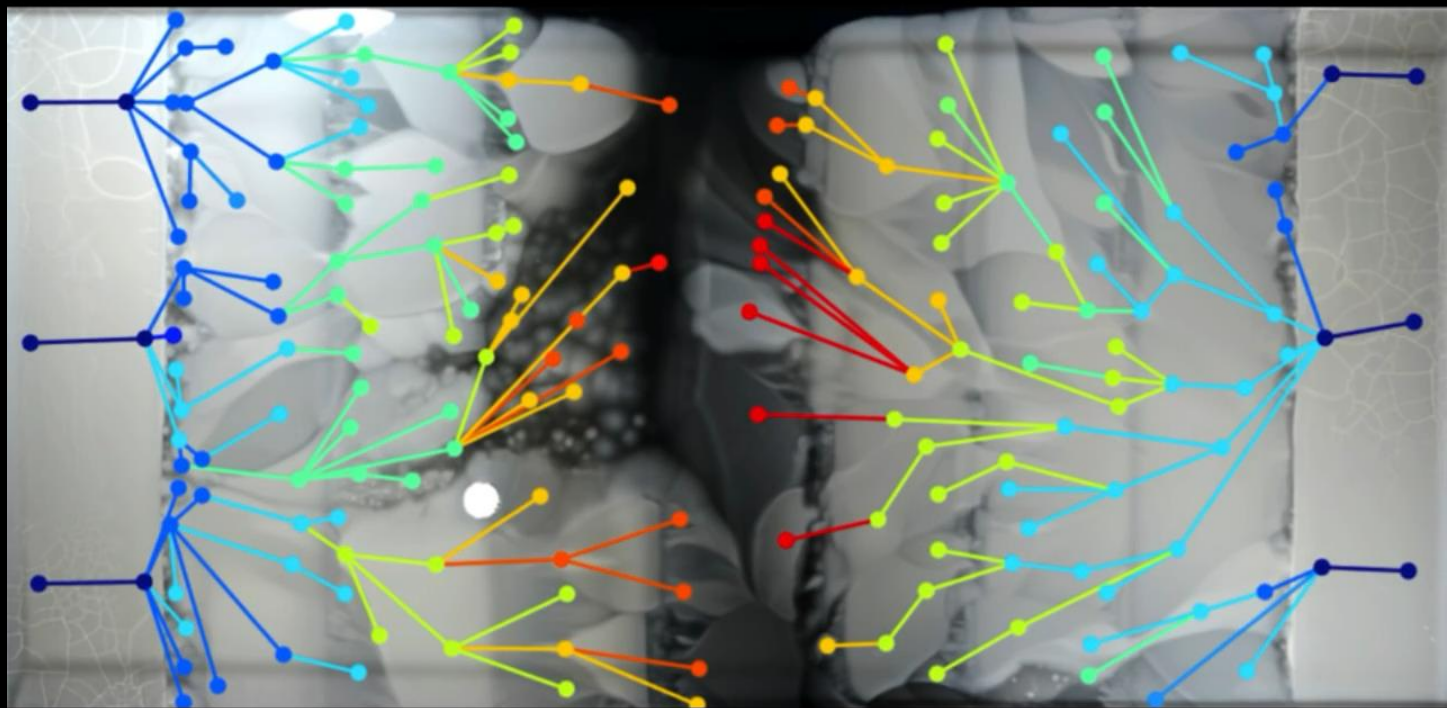


Evolution by natural selection

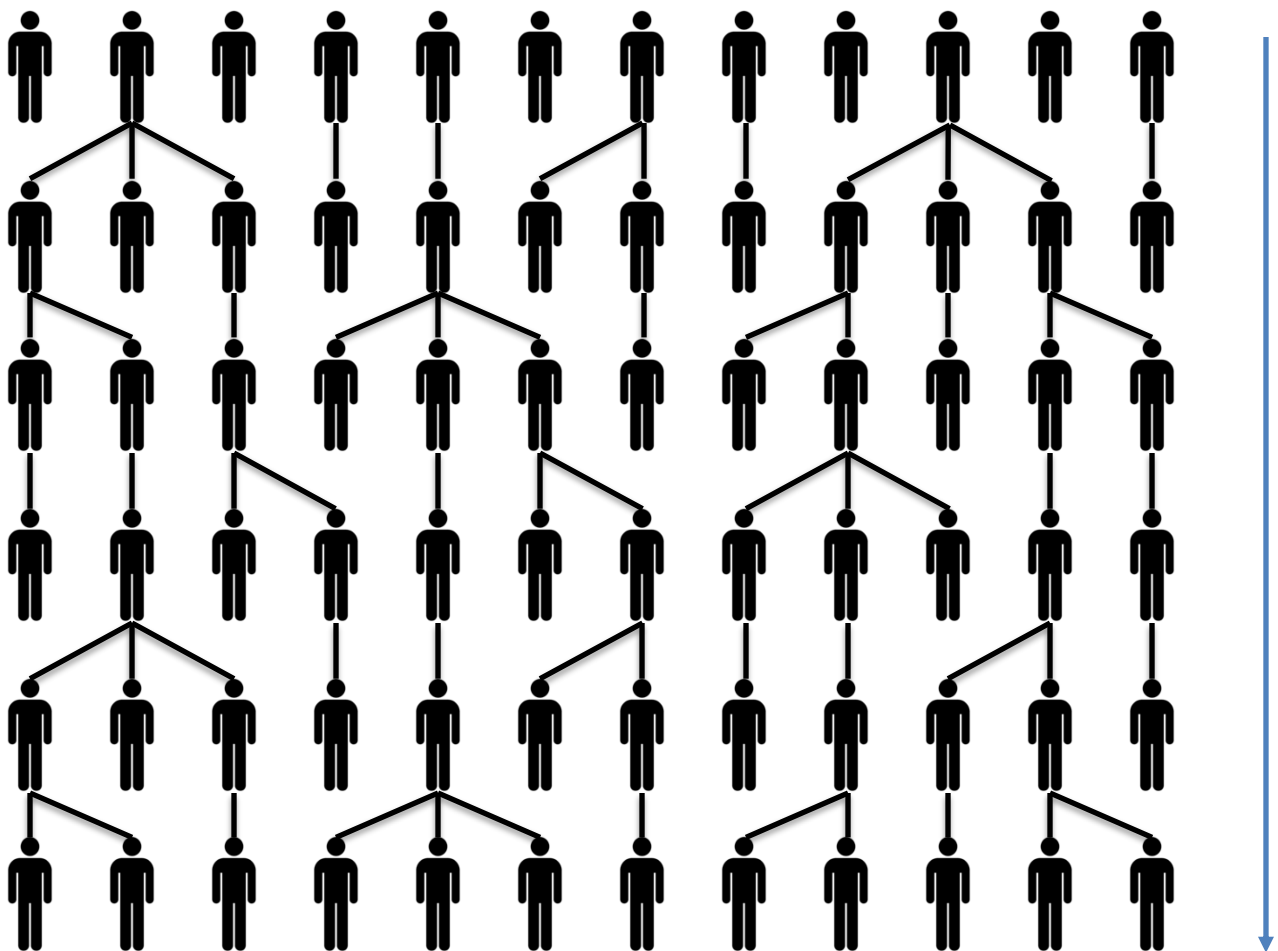
“Natural selection is easiest to understand, in the abstract, as a logical *argument*, leading from premises to conclusion...

1. Reproduction. Entities must reproduce to form a new generation.
2. Heredity. Offspring must tend to resemble their parents: roughly speaking, “like must produce like.”
3. Variation in individual characters among members of a population. ...
4. Variation in the *fitness* of *organisms* according to the state they have for a heritable character. ... individuals in the population with some characters must be more likely to reproduce (i.e. have higher fitness) than others

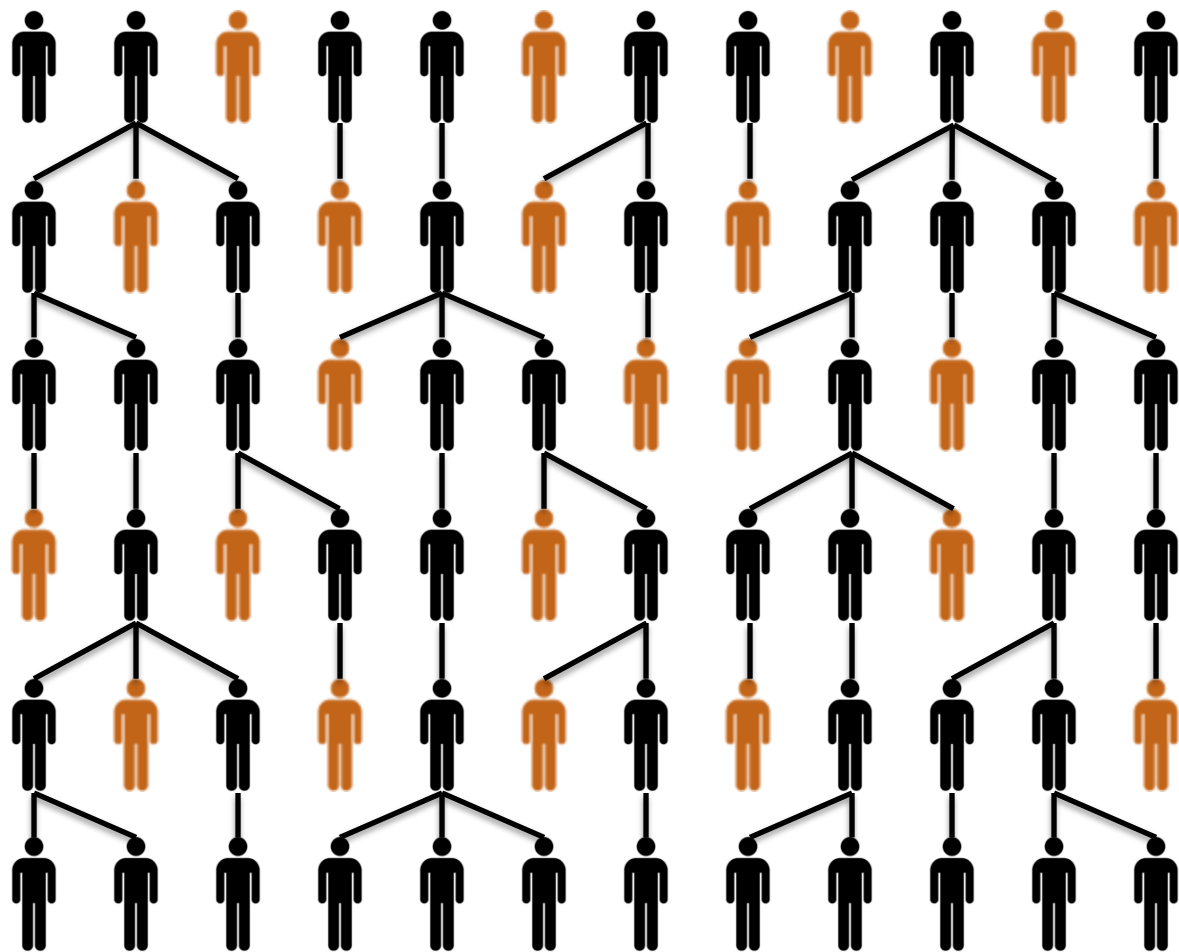
If these conditions are met, for any property or species, natural selection *automatically* results. If any conditions are not met, natural selection does not result. ... When all four conditions apply, the entities with the property conferring higher fitness will leave more offspring, and the frequency of that type of entity will increase in the population” (Ridley, *Evolution*, p. 71-72)

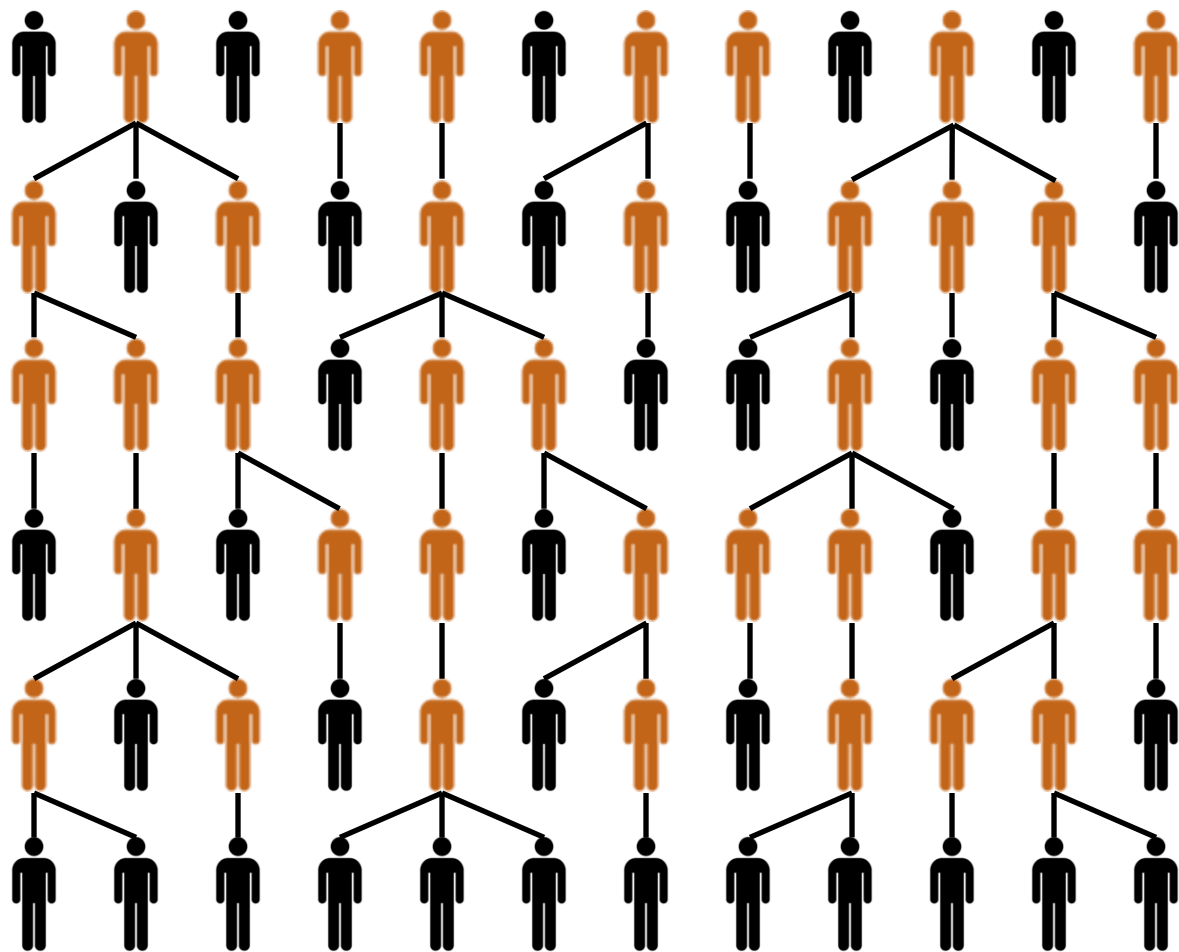


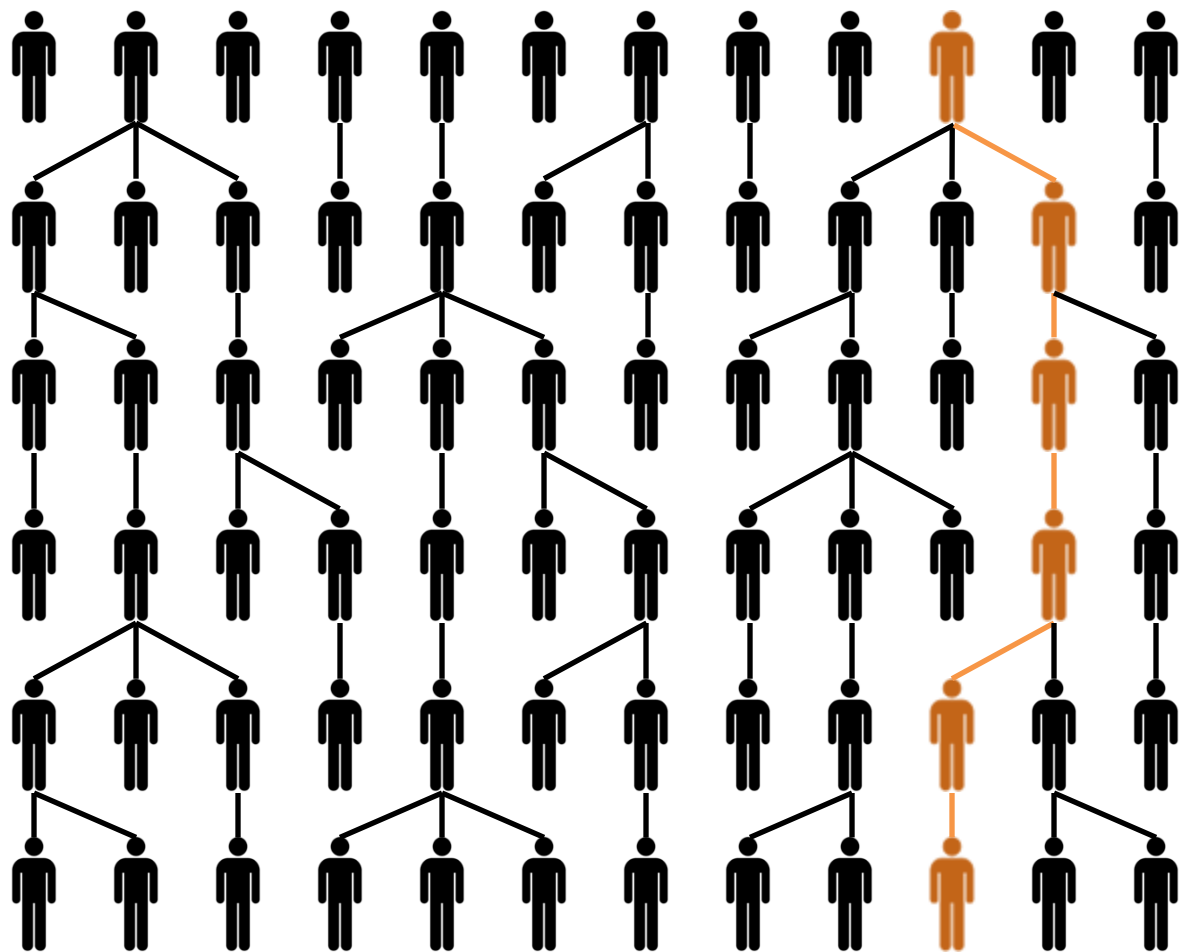
<https://www.youtube.com/watch?v=pIVk4NVIUh8>

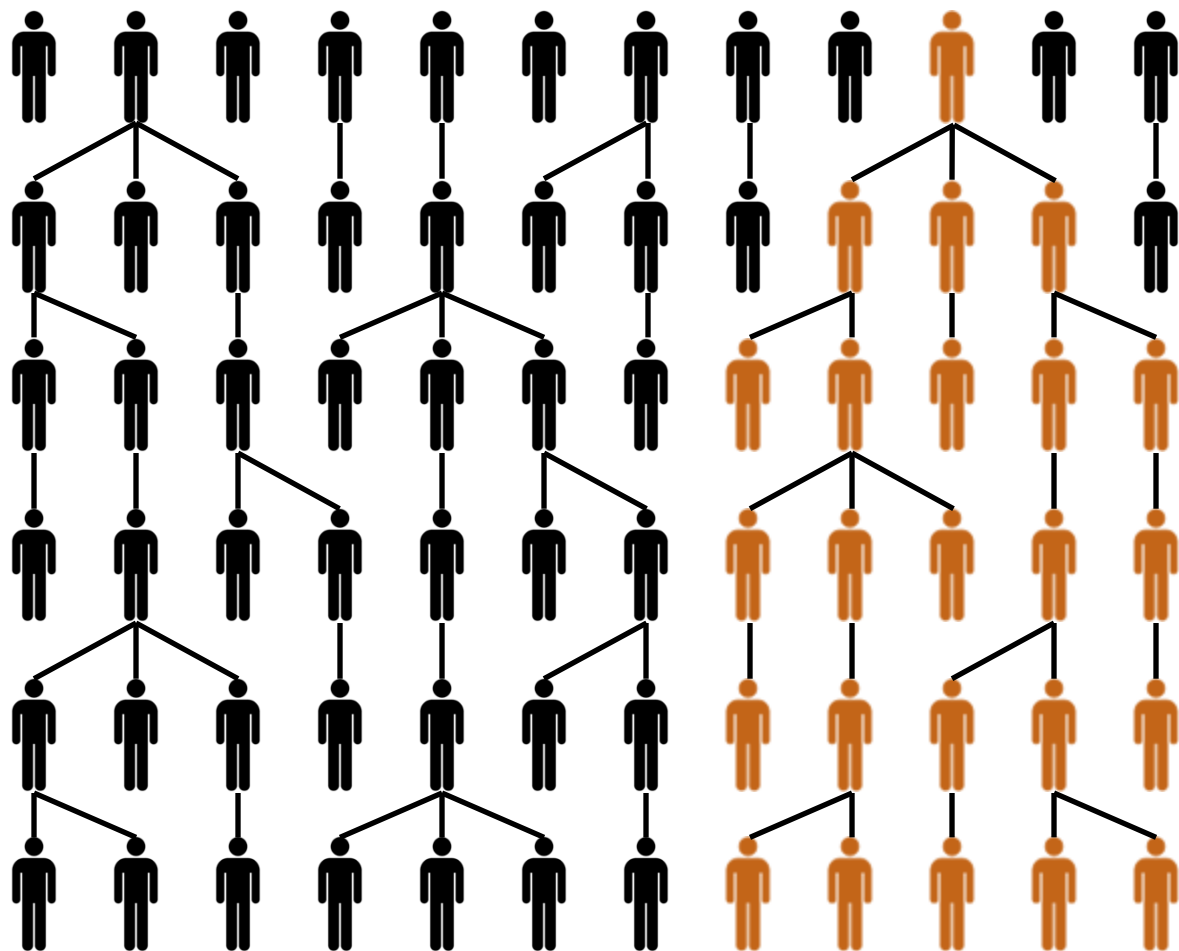


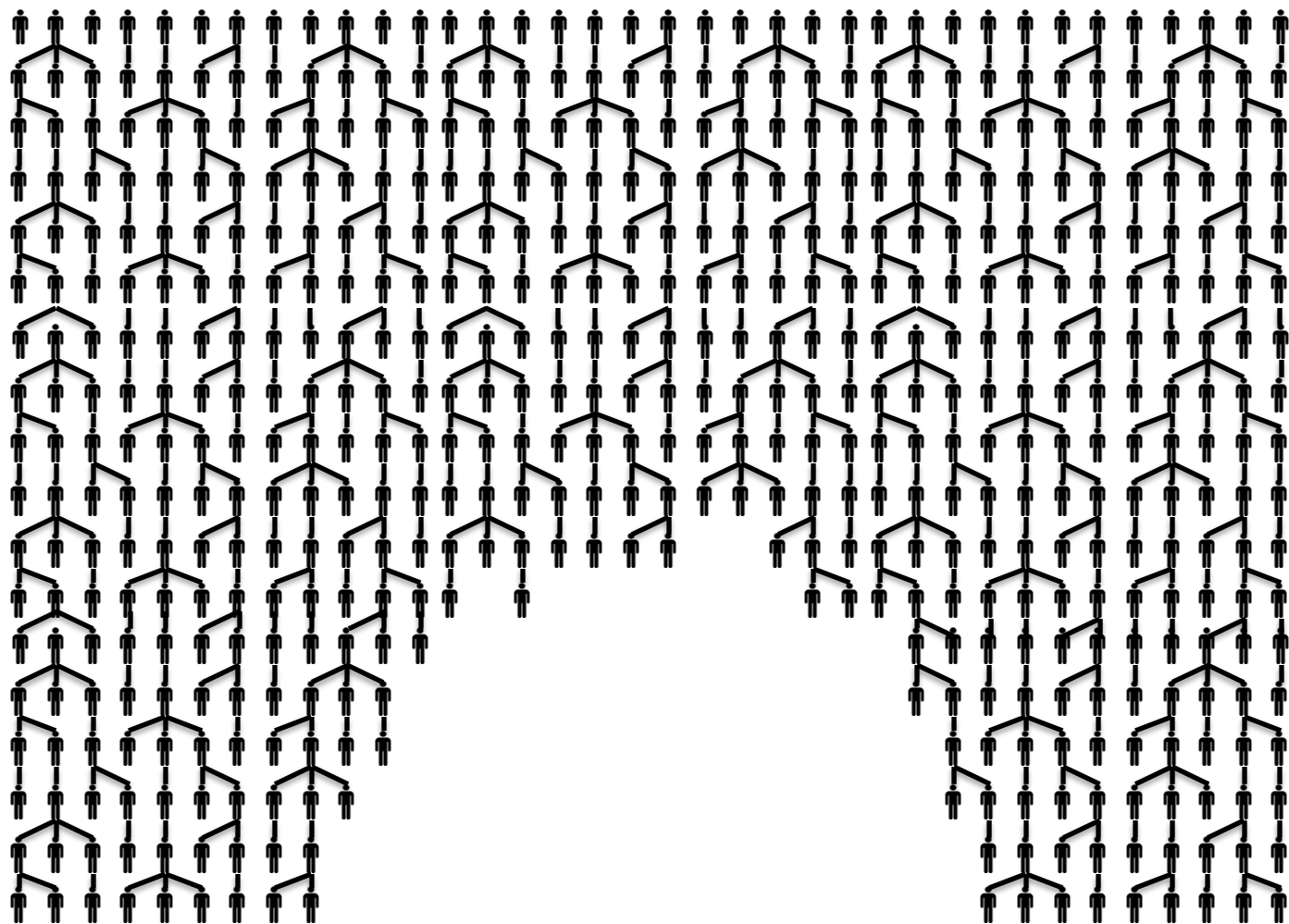
Time
(generations)

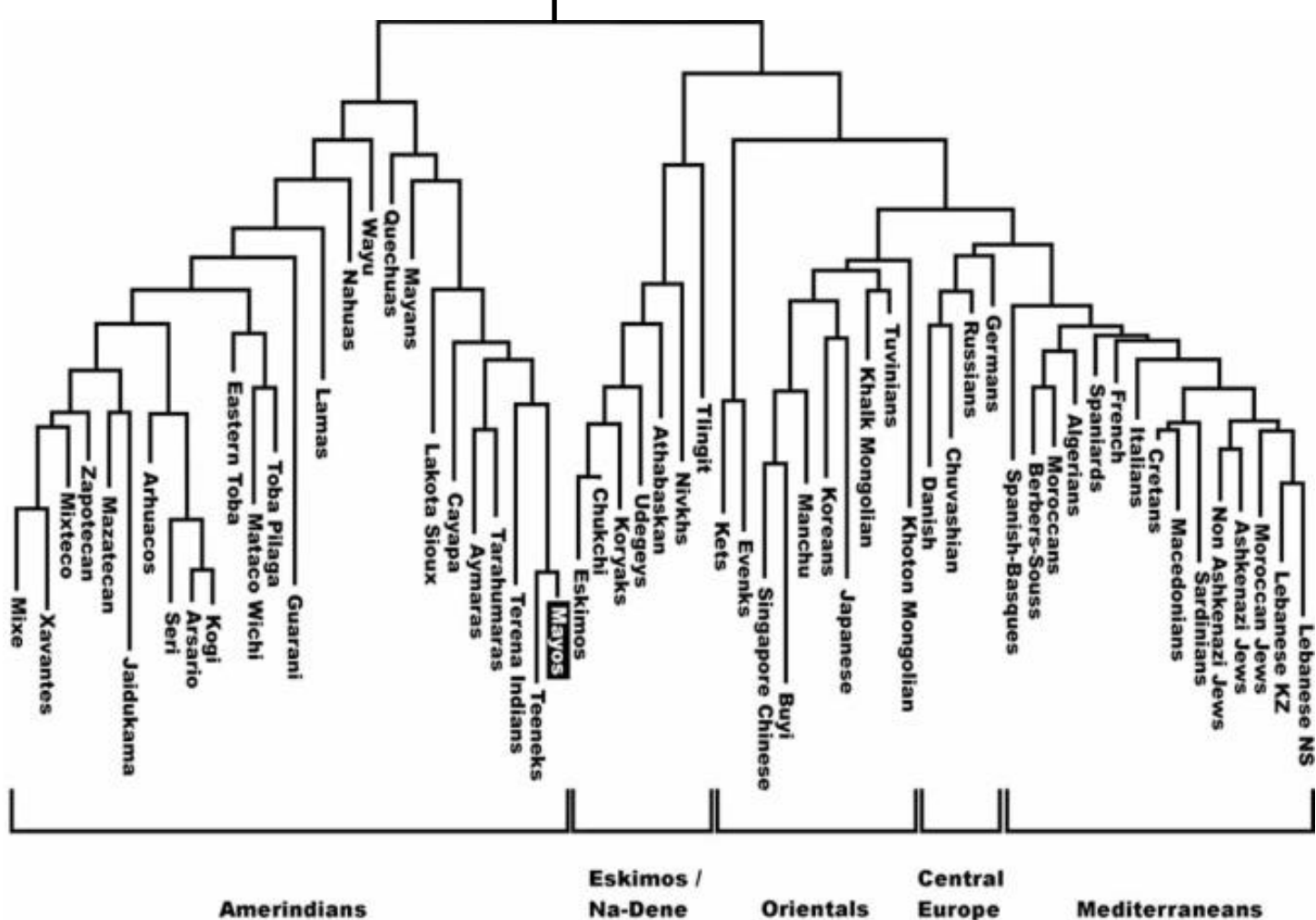


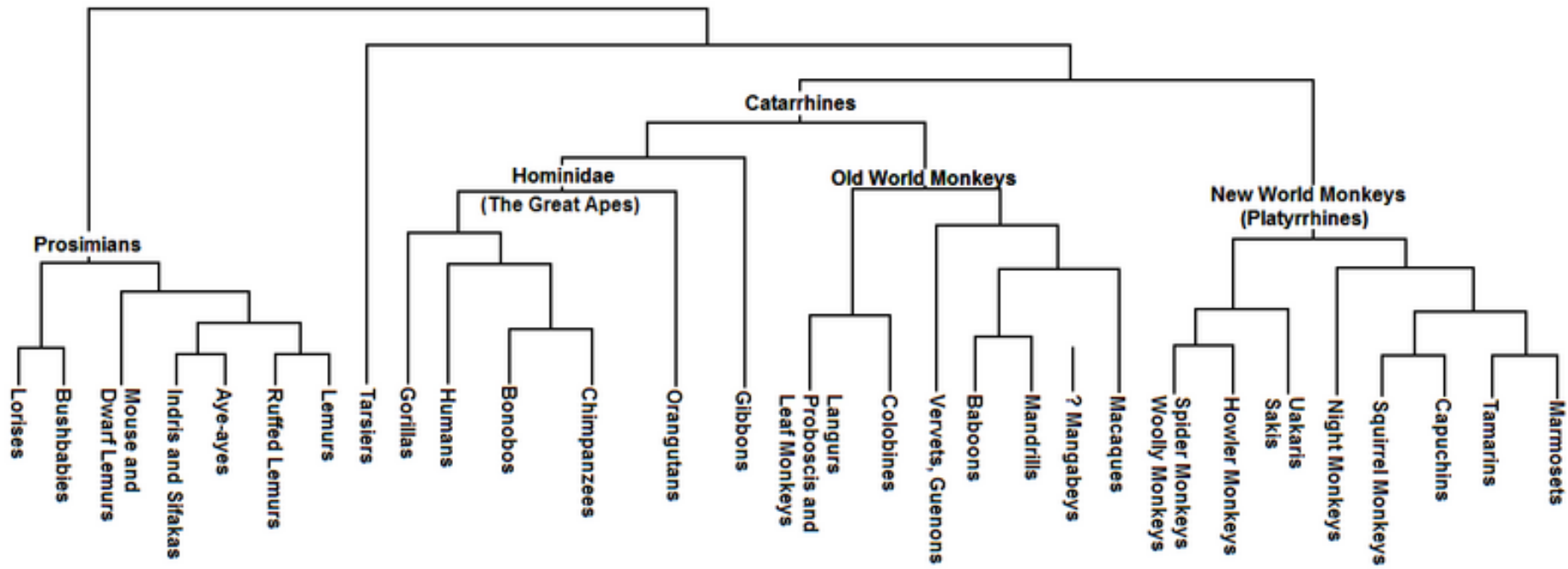


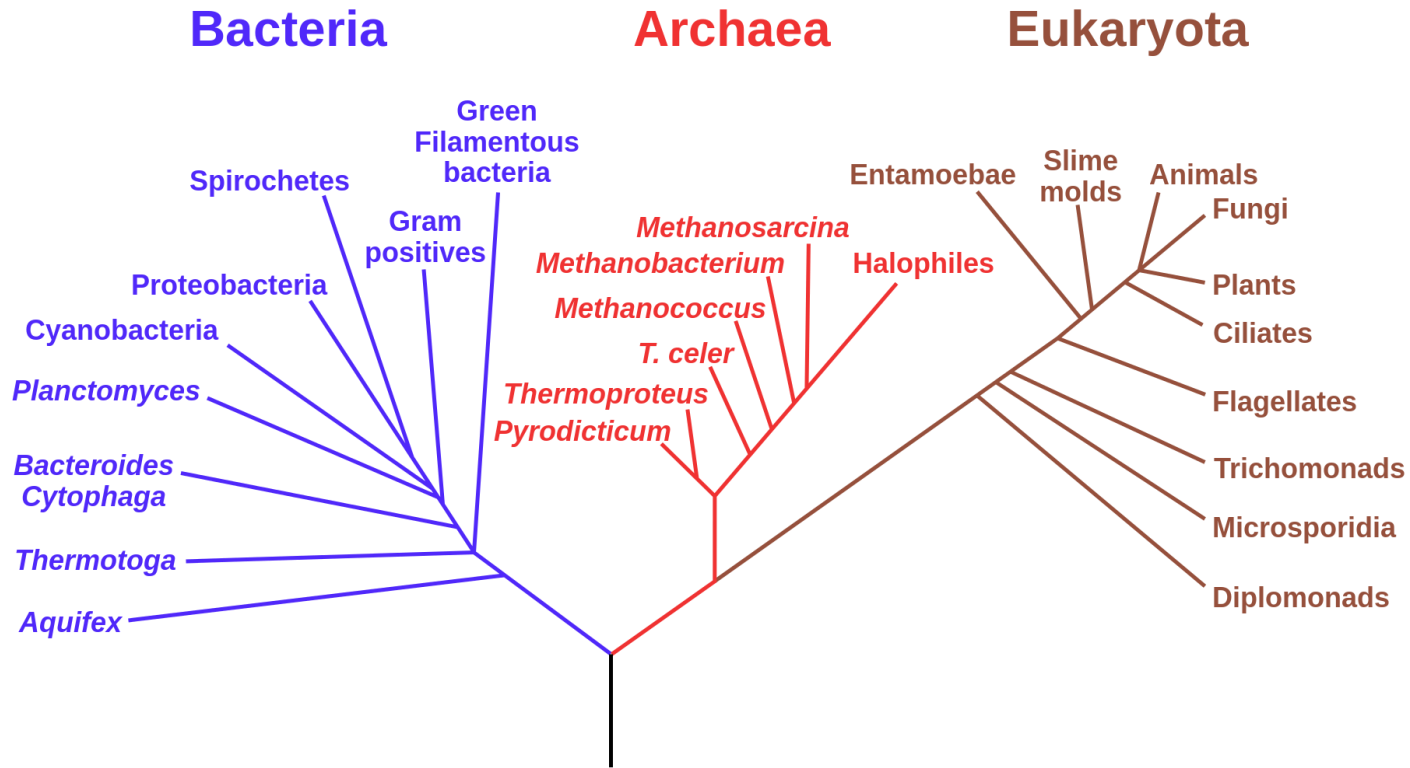




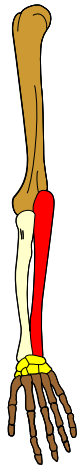




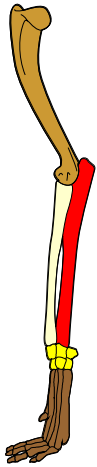




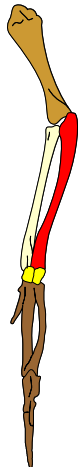
Homology



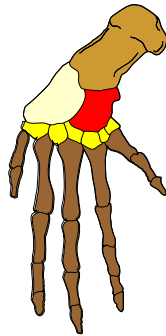
Human



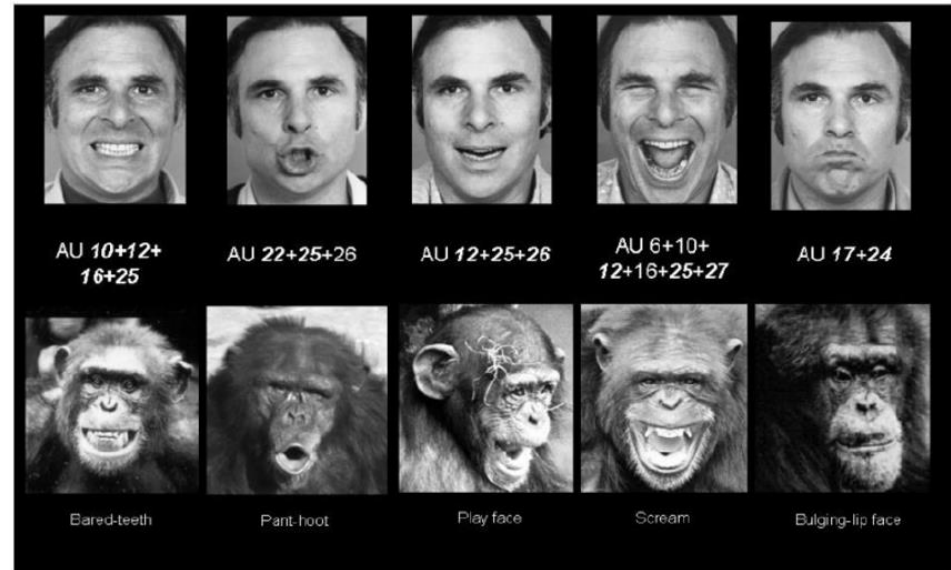
Dog



Bird

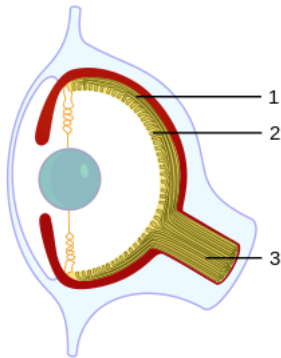
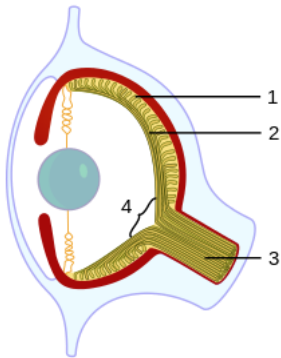


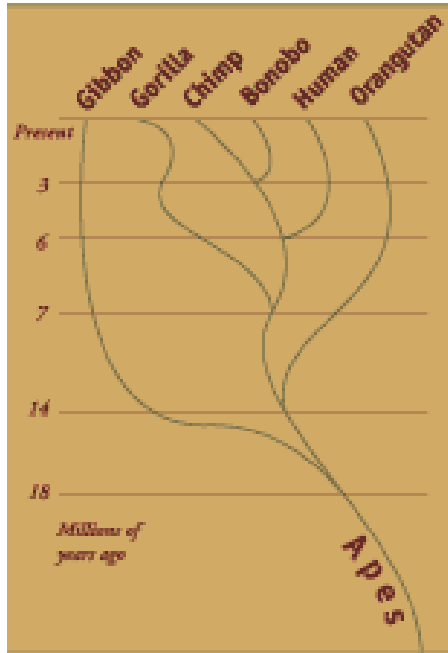
Whale



From Parr, L. A., & Waller, B. M. (2006). Understanding chimpanzee facial expression: insights into the evolution of communication. *SCAN*, 1, 221-228.

Convergent evolution





Humans and white-handed gibbons produce complex vocalisations featuring rule-governed combinations of a limited set of discrete sound types. Is this likely to be a homologous trait?

1: Yes

2: No

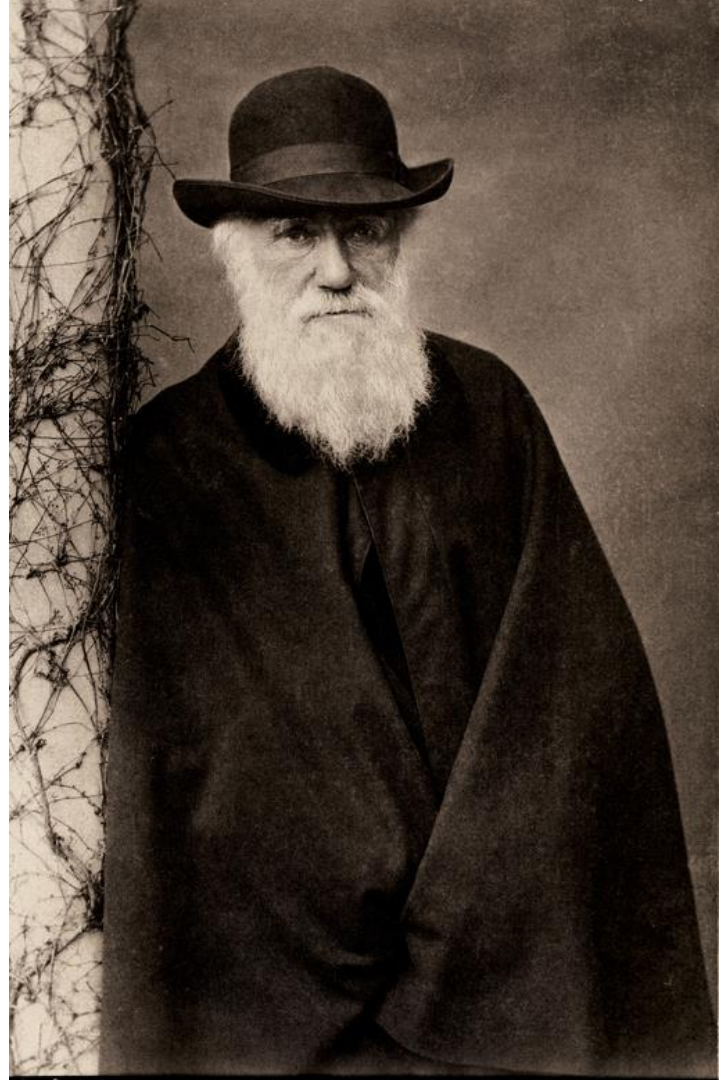
3: I need more information



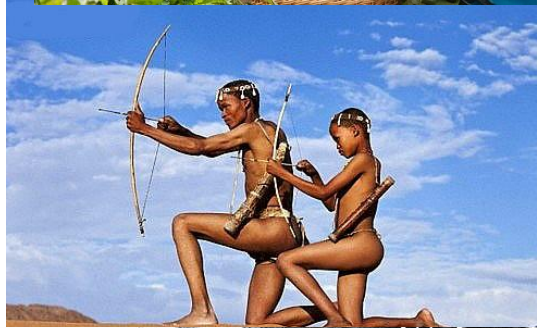
Language as an adaptation

Language as an adaptation?

- Language is just like any other **adaptive** feature of an organism's biology
- It's an **innate** feature of the human mind
- It **evolved by natural selection** under pressure for communication



A tool for the communication of knowledge and internal states



The core of the argument (from Pinker & Bloom, 1990)



“We will suggest that language shows signs of design for the communication of propositional structures over a serial channel.

The propositions in [human thought] are relational structures whose symbols pertain to people, objects, and events, the categories they belong to, their distribution in space and time, and their causal relations to one another ... The causal relations governing the behavior of other people are understood as involving their beliefs and desires, which can be reconsidered as relations between an individual and the proposition that represents the content of that belief or desire”

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.
Lengthy quote is from p. 712-713

The core of the argument (from Pinker & Bloom, 1990)



“We would want to be able to

- refer to individuals and classes,*
- to distinguish among basic ontological categories (things, events, places, times, manners, and so on),*
- to talk about events and states, distinguishing the participants in the event or state according to role (agents, patients, goals),*
- to talk about the intentional states of ourselves and others*
- to express distinctions of truth value, modality (necessity, possibility, probability, factivity)*

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.
Lengthy quote is from p. 712-713

The core of the argument (from Pinker & Bloom, 1990)



- *to comment on the time of an event or state, including both its distribution over time (continuous, iterative, punctate) and its overall time of occurrence*
- *to encode an unlimited number of predicates, arguments, and propositions*
- *to be able to use the same propositional content within different speech acts; for instance, as a question, a statement, or a command.*
- *to focus or to put into the background different parts of a proposition, so as to tie the speech act into its context of previously conveyed information and patterns of knowledge of the listener. ”*

The core of the argument (from Pinker & Bloom, 1990)



“All we have argued is that human language, like other specialized biological systems, evolved by natural selection. Our conclusion is based on two facts that we would think would be entirely uncontroversial: Language shows signs of complex design for the communication of propositional structures, and the only explanation for the origin of organs with complex design is the process of natural selection.” (p. 726)

Thoughts – are you persuaded, what evidence would be persuasive?

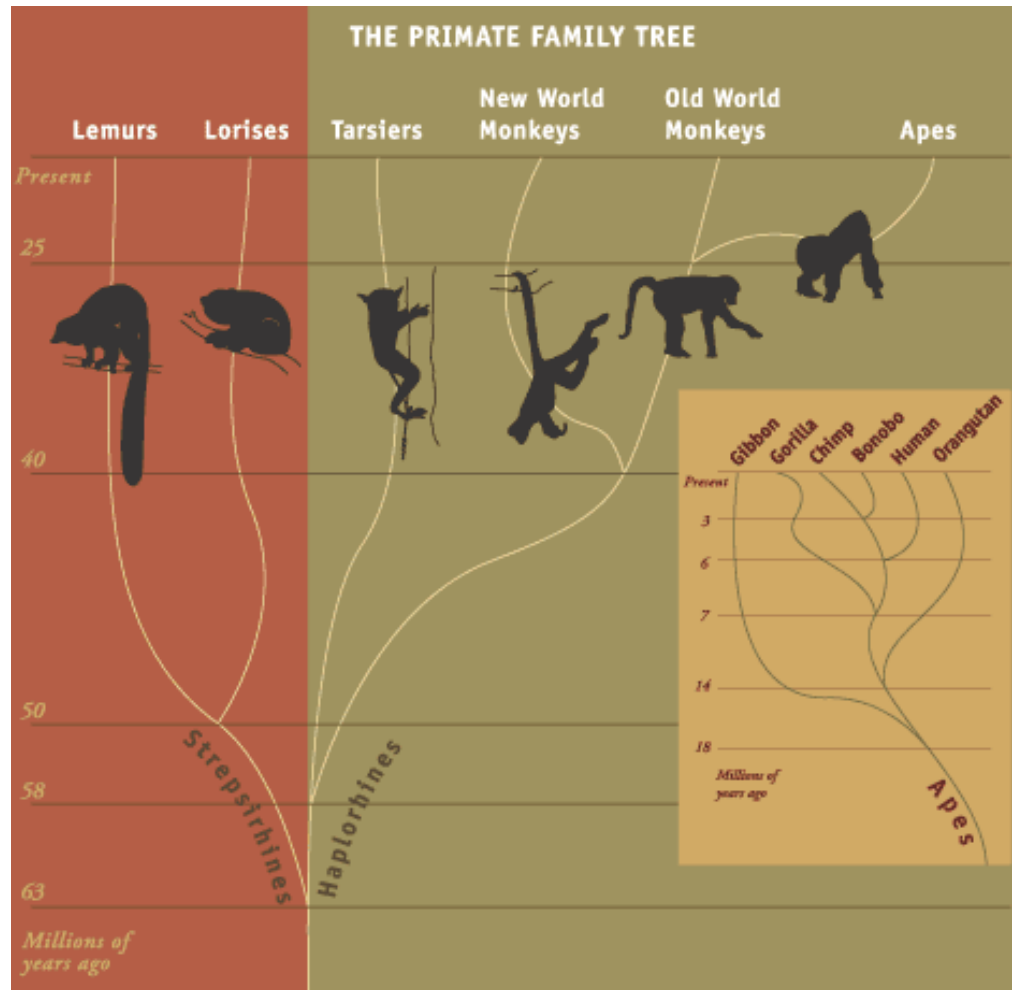
Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.

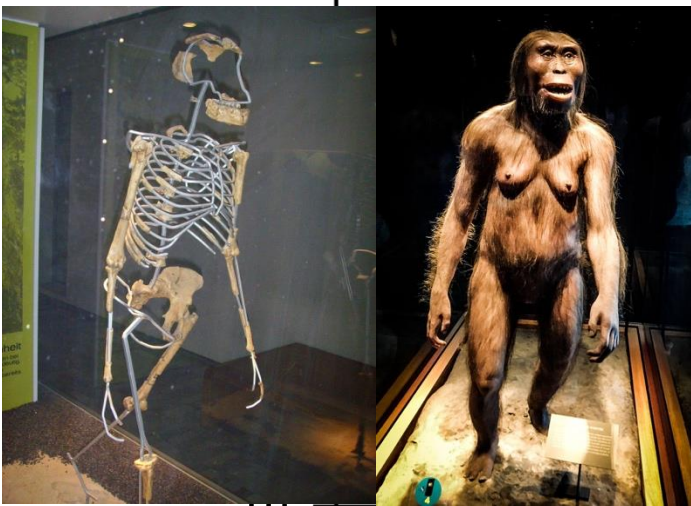
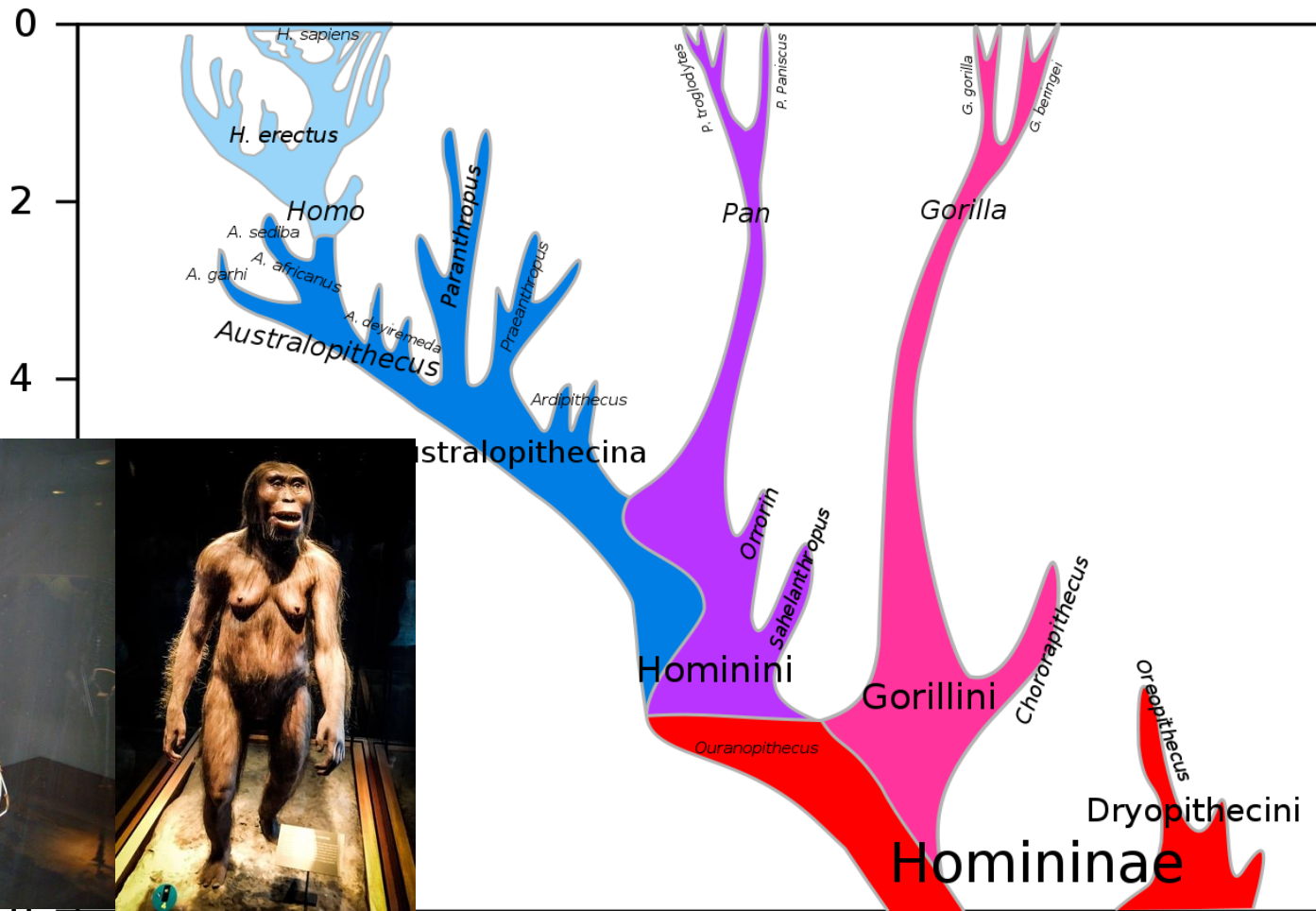
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Basic of human evolution



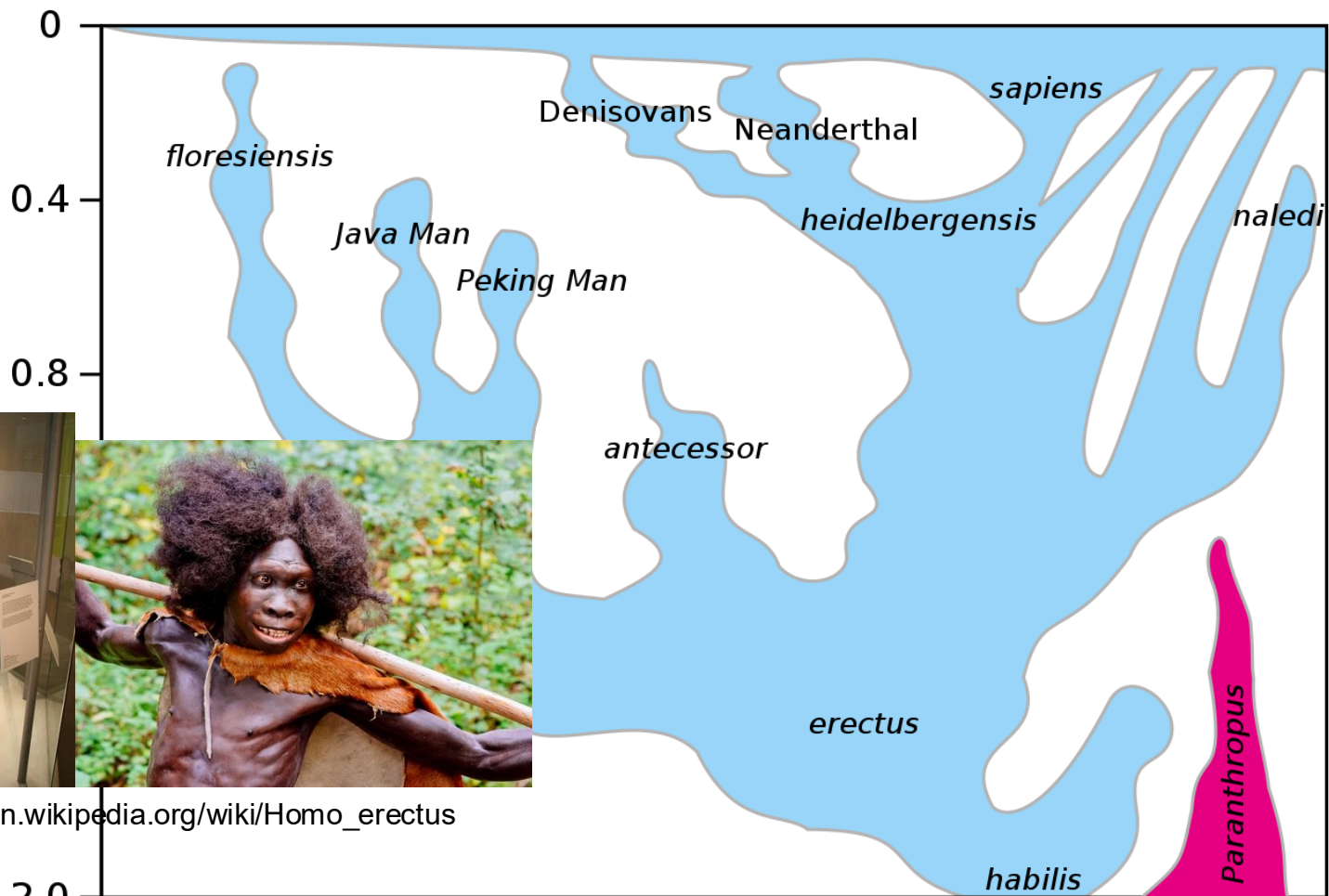


https://en.wikipedia.org/wiki/Lucy_%28Australopithecus%29

https://commons.wikimedia.org/wiki/File:Hominini_lineage.svg

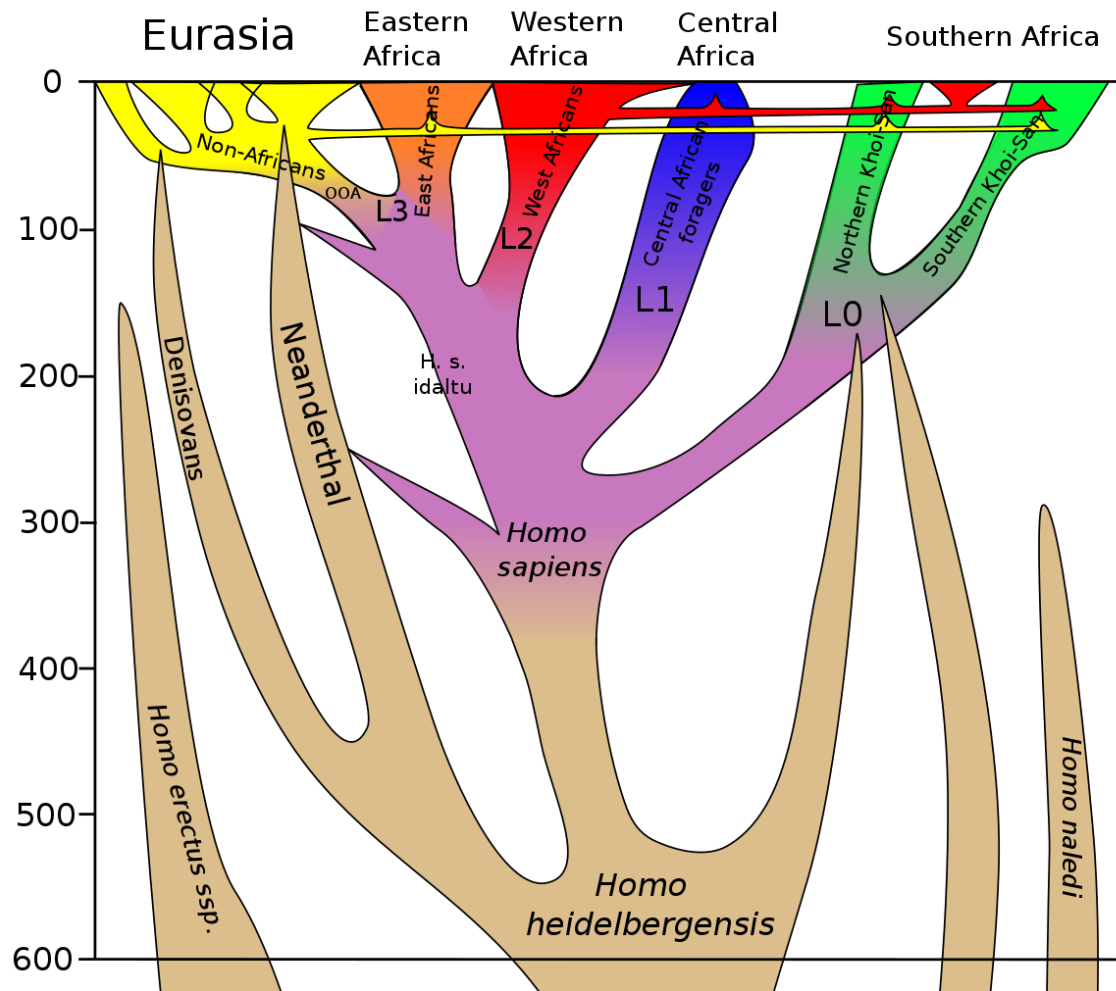
Eurasia

Africa

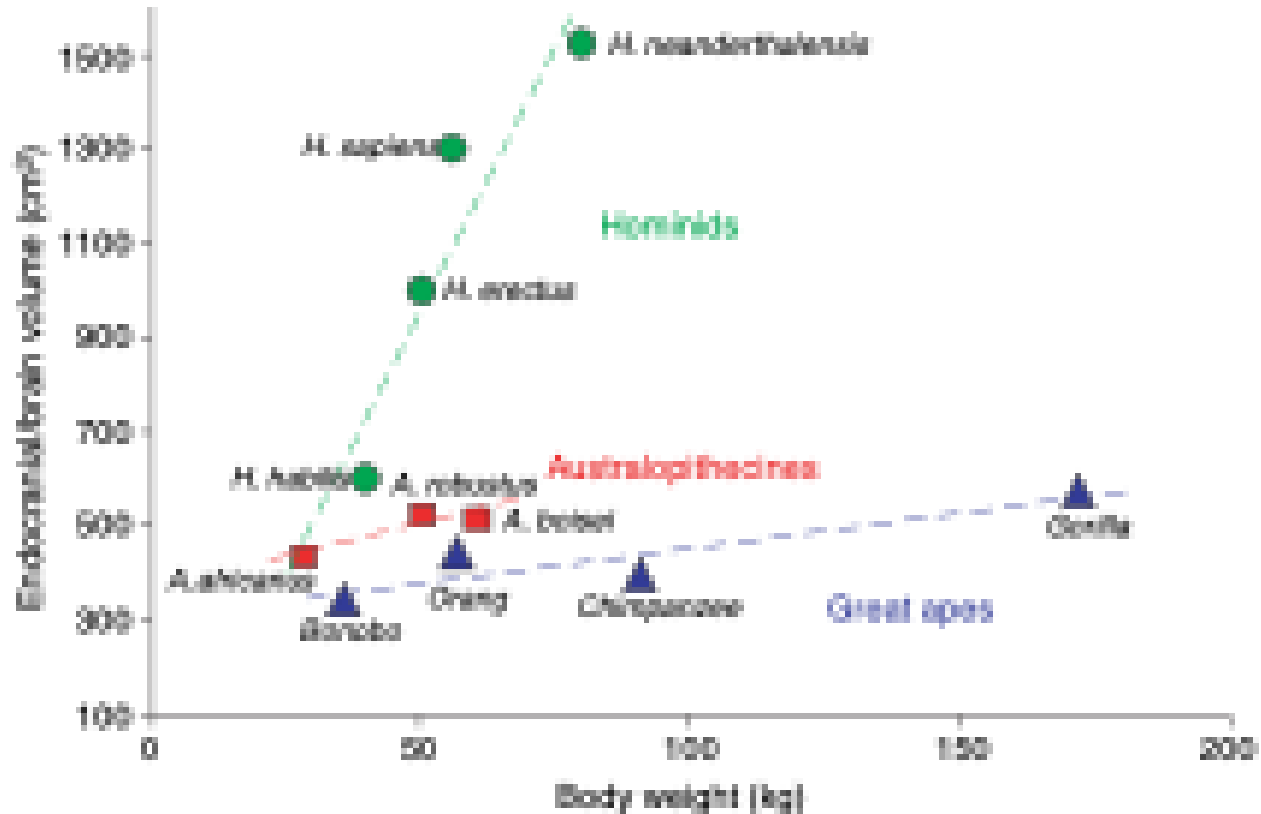


https://en.wikipedia.org/wiki/Homo_erectus

https://commons.wikimedia.org/wiki/File:Homo_lineage_2017update.svg



The evolution of brain size in great apes



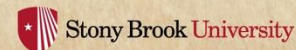
From Roth, G., & Dicke, U. (2005). Evolution of brain and intelligence. *Trends in Cognitive Sciences*, 9, 250-257.

DEVELOPMENT OF STONE TOOL TECHNOLOGY IN EASTERN AFRICA

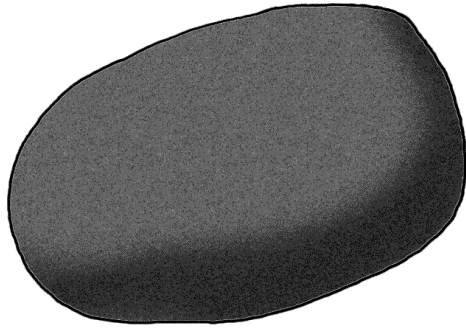
Location of
Lomekwi 3 site



Image Copyrights: Acheulean, Lomekwian tools © MPK-WTAP;
Oldowan tool © Turkana Basin Institute.



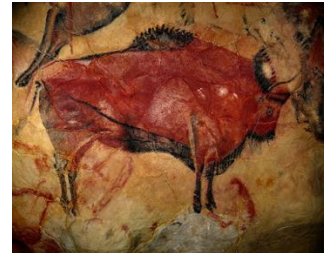
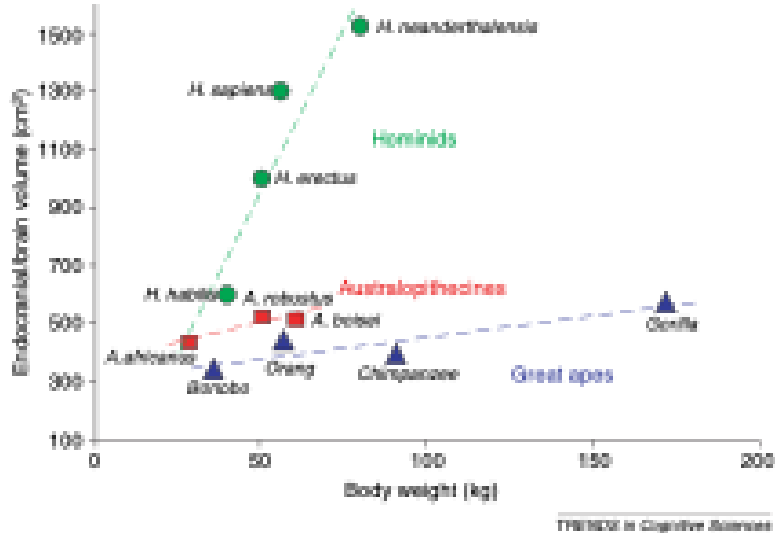
Mousterian
0.3 MYA – 40KYA



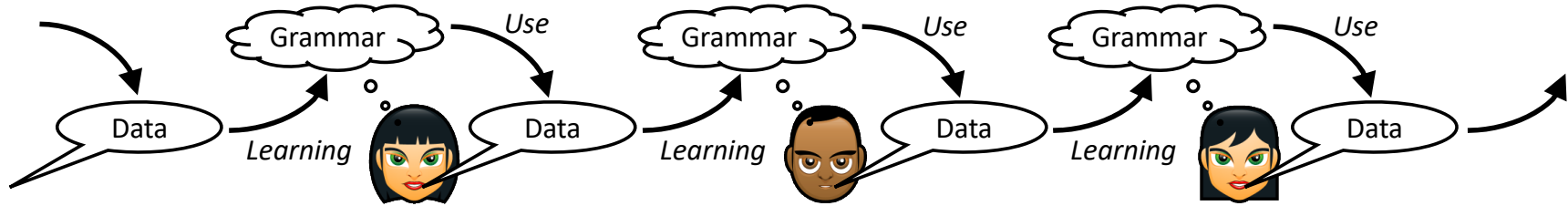
Upper Paleolithic Revolution
“Great Leap Forward”, 100-40KYA (?)



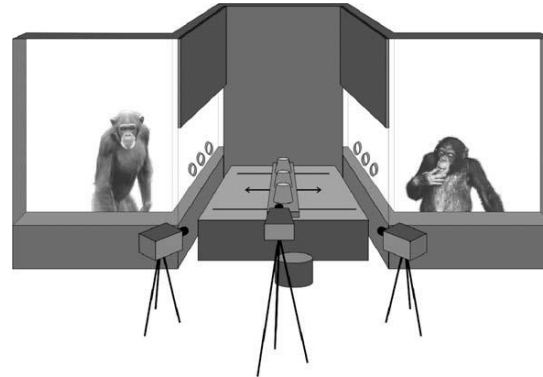
Are these informative about language?



Experiments looking at iterated learning



Experiments on learning and cognition in our closest relatives



The core of the argument (from Pinker & Bloom, 1990)



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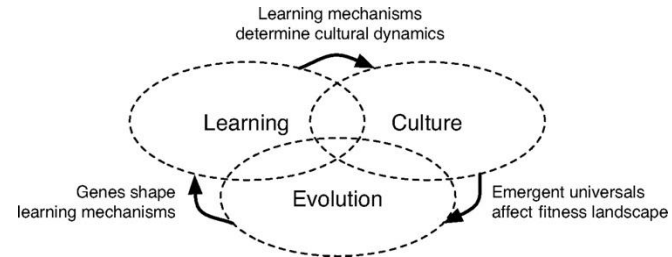
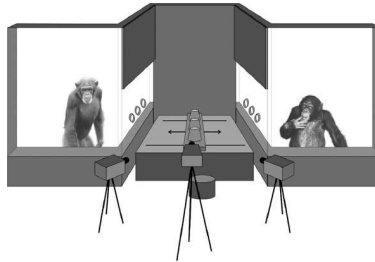
Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.

Next class

The alternative view: definitional structural properties of language (“design features”) are a product of cultural evolution, driven by biases in learning and use

- Symbols, compositionality, duality of patterning

We'll come back to biological evolution in classes 4-5



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